Supply Chain Black Box: Hidden Supplier Dependency Discovery, Calibrated Cascading-Risk Prediction and Optimised Recovery for an FMCG Manufacturer

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***Abstract*—A fast-moving consumer goods (FMCG) manufacturer of chocolate and malt beverages sees its Tier-1 suppliers but not the Tier-2/3 suppliers they share. In a documented *synthetic* supply network (500 suppliers, 69 % of links hidden) we test whether ranked estimates of hidden links improve recovery decisions. The system predicts supplier disruption, ranks candidate hidden links, simulates cascades and plans recovery with a mixed-integer linear program (MILP); two real datasets check transferable components only. Observed links understate Tier-2/3 exposure 2.1×/10.8×; logistic regression out-ranks the pre-specified XGBoost model; the calibration gain is not significant and graph neural networks did not beat tabular models. In 500 paired simulations of a severe outage of the five largest cocoa traders, the optimisation policy lowers simulated impact from 4.72 to 1.16 crore Indian rupees (INR). A factorial ablation attributes most of this to the predicted links: with the same MILP, the observed graph alone gives 3.64 crore. For mild shocks doing nothing is cheapest in all 15 sensitivity settings. All monetary results are simulated and assumption-conditional, not estimates of real-world savings.**

***Index Terms*—supply chain risk, link prediction, probability calibration, cascading disruption, recovery optimisation, ablation, FMCG**

I. Introduction

A. Business Problem

Disruptions are followed by significantly lower long-run stock returns [1]. Each Tier-1 input of a malt beverage (malt, milk powder, cocoa, sugar, packaging) depends on Tier-2 traders that procurement rarely maps, so shocks ripple through invisible links [2]. Fig. 1 shows the system that supports two decisions: *(D1) each month, which mitigation action to take for each supplier; (D2) when a shock is reported, which alternates receive which quantities, by which transport mode.*

B. Related Work, Novelty and Provenance

Search. A targeted (not systematic) search on 13 Nov 2026 of a web search engine, RePEc/IDEAS, SSRN and arXiv used the terms hidden-link prediction, deep-tier visibility, disruption mitigation optimisation and supplier disruption prediction. **Closest work** (Table I): link prediction recovers hidden supplier links [3], [4]; deep-tier *true* visibility improves simulated resilience assessment [5]; machine learning (ML) predicts first-tier disruption [6]; graph neural networks (GNNs) are benchmarked on SupplyGraph [7]; operations research/-management science (OR/MS) and ripple-effect work models mitigation and propagation [2], [8]; few AI studies reach the response stage [9]. **Differentiator.** We found no study that feeds *predicted* (uncertain) links into a recovery optimiser and separates their value from the optimiser's by comparing observed-graph, predicted-graph and true-graph arms under identical conditions. The contribution is systems integration plus a controlled synthetic decision experiment, not new methods or real-world savings. **Provenance** (author declaration): built in the public repository from 12 Nov 2026 (dated git history); no earlier project of the authors was reused.

TABLE I
Scope of related work. ✓ = primary focus; ○ = partial coverage or review; blank = not addressed. HL hidden-link prediction; RP risk prediction; CA calibration; CS cascade propagation; RO recovery optimisation; PE policy evaluation.

Work	HL	RP	CA	CS	RO	PE
Brintrup *et al.* 2018 [3]	✓					
Kosasih & Brintrup 2022 [4]	✓					
Brintrup *et al.* 2020 [6]		✓				
Wasi *et al.* 2024 [7]		✓				
Taghizadeh *et al.* 2021 [5]	○			✓		✓
Snyder *et al.* 2016 [8]				○	✓	○
Ivanov *et al.* 2019 [2]		○		✓		
Baryannis *et al.* 2019 [9]		○			○	
This work (synthetic)	✓	✓	✓	✓	✓	✓

C. Research Questions, Hypotheses and Map

RQ1/H1: point-in-time features give above-chance 30-day risk ranking, and calibration lowers ECE (expected calibration error). **RQ2/H2**: the observed network understates upstream exposure, and link prediction ranks hidden links above the base rate in average precision (AP). **RQ3/H3**: GNNs beat tabular models. **RQ4a/H4a**: the optimisation policy C has lower expected impact than rule-based B and no intervention A *at every severity*. **RQ4b/H4b**: holding the MILP fixed, adding

TABLE II
RESEARCH MAP. E3 IS A SEPARATE SHIPMENT-LATENESS TASK ON REAL DATA; IT DOES NOT VALIDATE THE SUPPLIER-RISK MODEL.

RQ	Exp. (data)	Test	Verdict	Output used in
1	E1 (synthetic)	bootstrap	partly	D1 action; MILP screen
2	E2A (synthetic)	AP vs. base	supported	exposure; C's graph
3	E2B (synth.+real)	5 seeds	not supported	model choice
–	E3 (DataCo, real)	bootstrap	transfer only	lateness API
4a	S1–S9 (simulated)	Wilcoxon–Holm	not supported	tiered policy
4b	Ablation (simul.)	Wilcoxon–Holm	severe only	novelty claim

predicted links lowers impact. Table II links each question to its experiment and verdict.

II. METHODOLOGY

A. Data, Units and Provenance

Synthetic ecosystem (fictional, seed 42): 120/200/180 Tier-1/2/3 suppliers, 50 materials, 100 products, two plants, 1,048 disruptions (782 qualifying) in 2024–2025. A buyer of material m picks $k \in \{2,\dots,6\}$ sources with probability $\propto \text{size}^{1.6} e^{-d/2500\,\text{km}}$; Tier-2/3 links are visible with probability 0.45/0.15 (461 of 1,500 observed). Disruption hazards include contagion over the *true* network; hidden links serve only as ground truth. **DataCo** [19] (real, CC BY 4.0; the Mendeley V5 file obtained from a pinned public mirror and SHA-256 checked): 180,519 order lines, 7,754 cancelled removed; it has no supplier identities. **SupplyGraph** [7] (real FMCG): 40 products, 684 edges, 221 days. The synthetic tables pass 178 automated validation checks. Money is INR; 1 lakh (L) = 10^5 and 1 crore (Cr) = 10^7 INR.

B. Disruption Prediction and Leakage Control (RQ1)

The prediction timestamp t is 00:00 on the first of each month. For supplier i,

$$y_i(t) = \mathbb{1}\big[\exists e:\ s_e \in (t, t+30\text{d}],\ \ell_e \geq 0.25,\ d_e \geq 3\big], \quad (1)$$

with start s_e, capacity loss ℓ_e and duration d_e (days). The 25 features use only information available before t and *observed* links. Splits: train 2024-04..2025-02 (5,500 supplier-months), one-month embargo, calibration 2025-04..06 (1,500), test 2025-07..12 (3,000). Logistic regression (LR), a decision tree, a random forest and XGBoost [10] are compared by the area under the ROC curve (ROC-AUC), the area under the precision–recall curve (PR-AUC), the Brier score and ECE; Platt scaling [11] and isotonic regression [12], [13] are fitted on the calibration split; the method is chosen by 5-fold cross-validated Brier score *within* that split, never on test data. XGBoost was fixed as the primary model in the code before any result was seen; its Platt-calibrated probability q_i feeds D1, the MILP risk screen and the simulated failure chance of alternates.

C. Link Prediction and Graph Models (RQ2, RQ3)

The graph has 501 nodes: 500 suppliers and the manufacturer; supplier-level results use the 500 suppliers. *Link prediction* scores candidate pairs; *hidden-link discovery* evaluates those scores against the synthetic ground truth. **A1**: observed links split 70/10/20 with test links removed from message passing. Comparators: heuristics [14], node2vec [15], a multilayer perceptron (MLP) and GNNs. **A2**: rank all 8,863 material-compatible unobserved pairs against 1,039 hidden links; the model is chosen by validation AP. **E2B** feeds E1 features through GCN [16], GraphSAGE [17] and GAT [18] on observed, observed+top-500 predicted and true graphs, and predicts a 7-day shortfall on SupplyGraph.

D. Exposure, Simulation and Decision Rule (D1)

Exposure E_i (INR) is the expected contribution-margin loss at the manufacturer if supplier i is disrupted (CM = contribution margin):

$$E_i = \textstyle\sum_m r_{im}\,\text{CM}_m\,\mathbb{E}\big[S\max(0, D - c_m - 10(\tau_i - 1))\big], \quad (2)$$

with reach r_{im} (product of dependency shares along paths), daily contribution margin CM_m, cover c_m (days), tier τ_i and empirical severity/duration (S, D). The simulator propagates availability daily over the true network. The action is $a^* = \arg\min_a C_a + q_i(1-\rho_a)E_i$ over four actions with cost C_a = 0/5k/20k/60k INR and assumed risk reduction ρ_a = 0/0.15/0.45/0.70; it is Bayes-optimal only if q_i is calibrated, with break-even $q^* = \Delta C/(E_i \Delta\rho)$.

E. Recovery MILP, Scenarios and Robustness Design (RQ4)

For deficits δ_m the MILP (PuLP/CBC) chooses quantities x_{jk} from candidate j by mode k, onboarding z_j and shortage s_m:

$$\min\ \textstyle\sum_{jk} p_j(1 + f_{jk} + hW/2)x_{jk} + \sum_j F z_j + \sum_m \pi_m s_m \quad (3)$$

$$\text{s.t. } \textstyle\sum_{jk} \rho_j(1-q_j)\kappa_j\alpha_j x_{jk} + s_m \geq \delta_m,$$
$$x_{jk} \leq u_j(W - L_{jk})^+,\ \ \textstyle\sum_k x_{jk} \leq 0.6\,\delta_m,\ \ q_j \leq 0.35,$$

with price p_j, freight f_{jk}, holding rate h, deadline W, onboarding cost F, penalty π_m, compatibility κ_j, spare capacity u_j and availability α_j de-rated by estimated dependency on the disrupted suppliers. The shock removes 25/50/100 % of the capacity of the five largest Tier-2 cocoa traders for a nominal 60 days. **A** does nothing; **B** uses the observed graph, excludes any alternate linked to the failed traders and allocates by reliability when cover falls below 10 days; **C** uses the observed+predicted graph and the MILP at detection plus the same trigger. All arms share network, demand, inventory and illustrative economics, and replication r uses seed $s_0 + r$ for every arm (common random numbers, CRN; 180-day horizon). **Ablation (RQ4b):** a 2×2 factorial {rule, MILP}$\times${observed, observed+predicted graph} plus an oracle arm (MILP, true graph) isolates the value of predicted links. **Sensitivity:** one at a time, we vary management overhead, shortage penalty, interruption cost, duration, alternates' spare capacity and failure risk, and how many links the buyer knows (200 CRN replications per cell).

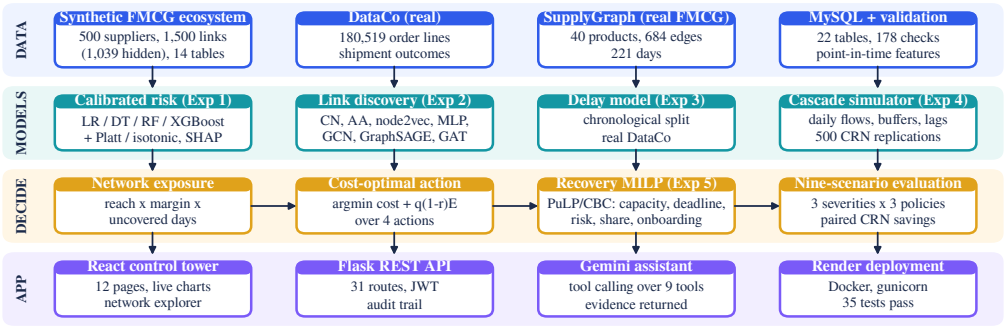


Fig. 1. System architecture. Real and synthetic data feed point-in-time features; risk, link, delay and cascade models feed an exposure model, a cost-based action rule and a recovery MILP, served by a Flask API, a React control tower and a tool-calling Gemini assistant (Render).

TABLE III
E1: 30-DAY SUPPLIER DISRUPTION PREDICTION ON THE UNTOUCHED TEST SPLIT (3,000 SUPPLIER-MONTHS, PREVALENCE 7.4%). AUCS AFTER PLATT SCALING; CALIBRATORS FITTED ON THE CALIBRATION SPLIT; BOLD = LOWEST ECE; SHADED = PRE-SPECIFIED MODEL.

			Brier		ECE_{10}		
Model	ROC-AUC	PR-AUC	raw	Platt	raw	Platt	Iso.
Logistic reg.	0.666	0.150	0.0669	0.0670	0.0066	**0.0059**	0.0070
Decision Tree	0.571	0.098	0.0704	0.0689	0.0366	**0.0111**	0.0124
Random Forest	0.657	0.140	0.0672	0.0675	**0.0046**	0.0081	0.0063
XGBoost	0.626	0.122	0.0689	0.0680	0.0197	**0.0104**	0.0109

TABLE IV
STATISTICAL TESTS. E1: BOOTSTRAP OVER 500 SUPPLIERS; E3: OVER 16,840 ORDERS. POLICIES: SAVING = IMPACT(FIRST)−IMPACT(SECOND) OVER 500 CRN PAIRS, HOLM OVER 9; CI OF THE MEAN; MED. = MEDIAN; WIN = % OF REPLICATIONS WHERE THE SECOND POLICY IS CHEAPER.

	Comparison	Mean	95% CI	Med.	Win%	p	p_{Holm}
Models: bootstrap, B=1,000, two-sided							
E1	LR−XGB, ROC-AUC	+0.040	[0.013, 0.068]	–	–	0.008	–
E1	LR−XGB, PR-AUC	+0.026	[0.002, 0.055]	–	–	0.044	–
E1	ECE, raw−Platt	+0.0095	[-0.002, 0.020]	–	–	0.108	–
E3	XGB−mode, AUC	+0.035	[0.030, 0.041]	–	–	<0.001	–
Policies: saving in INR lakh; two-sided Wilcoxon							
Mil.	A vs. B	-0.00	[-0.003, 0.000]	+0.00	0	0.109	0.167
Mil.	A vs. C	-0.82	[-0.823, -0.817]	-0.83	0	<0.001	<0.001
Mil.	B vs. C	-0.82	[-0.822, -0.816]	-0.83	0	<0.001	<0.001
Mod.	A vs. B	+11.0	[8.6, 13.6]	+0.00	18	0.083	0.167
Mod.	A vs. C	+16.7	[9.3, 24.7]	-11.8	17	<0.001	<0.001
Mod.	B vs. C	+5.73	[0.3, 11.4]	-11.7	10	<0.001	<0.001
Sev.	A vs. B	+74.1	[70.8, 77.4]	+93.3	83	<0.001	<0.001
Sev.	A vs. C	+356.2	[322.1, 386.4]	+283.4	80	<0.001	<0.001
Sev.	B vs. C	+282.0	[254.7, 310.9]	+186.0	73	<0.001	<0.001

F. Statistical Analysis

Bootstraps (B=1,000, percentile confidence intervals (CIs), two-sided) resample suppliers in E1 (their months are dependent) and orders in E3. Policies: the estimand is the mean paired saving Δ = impact(first) − impact(second), so $\Delta > 0$ means the second arm is cheaper; we report a bootstrap CI of the mean, the median, the share of replications in which the second arm is cheaper, and a two-sided Wilcoxon signed-rank test, Holm-adjusted within each family (9 policy comparisons; 12 ablation comparisons). α=0.05. Policies, shock, 13 outcome metrics, CRN and replication count were fixed before any scenario run.

III. RESULTS

A. Disruption Prediction (RQ1: partly supported)

Ranking is modest (prevalence 7.4 %; Table III, Fig. 2). Logistic regression is best (ROC-AUC 0.666, 95 % CI 0.626–0.705; PR-AUC 0.150) and significantly better than XGBoost (0.626, CI 0.587–0.664; Table IV); validation data also favoured it. We keep XGBoost to avoid test-set model choice; using logistic-regression risk moves severe C impact only from 1.16 to 1.17 Cr. Platt scaling lowered XGBoost's ECE from 0.020 to 0.010, not significantly (p=0.11); it helped the decision tree (0.037→0.011) and worsened the already-calibrated random forest. D1 probabilities rank above chance but are not shown to be operationally calibrated. On the test months the cost rule raised 5 alerts, none followed by a qualifying disruption; realised cost (118.3 L) was close to monitor-only (116.9 L). This is a small retrospective demonstration, not validation.

B. Exposure Gap and Link Prediction (RQ2: supported; RQ3: not supported)

Observed links understate Tier-2/3 exposure 2.1×/10.8× (Fig. 3). This measures exposure, not prediction. The validation-selected MLP pair model reaches AP 0.227 (mean of five seeds) against a base rate of 0.117 (precision 0.32 in its top 100; Table V); the single ranked list passed to the simulator (seed 0) has AP 0.220. This is useful ranking inside this synthetic candidate set, not proof that recovered links are real suppliers. GNNs did not improve risk classification: on synthetic data the best GNN reaches PR-AUC 0.111±0.011 (observed+predicted graph) and 0.108 with the true graph vs. 0.122 for XGBoost; on real SupplyGraph logistic regression (0.488) beats the best GNN (0.461±0.017; 5 seeds).

C. Transfer to Real Shipments (DataCo)

On 34,554 chronologically held-out order lines (16,840 orders) XGBoost reaches ROC-AUC 0.780 (CI 0.771–0.788; PR-AUC 0.858) vs. 0.744 for a shipping-mode baseline

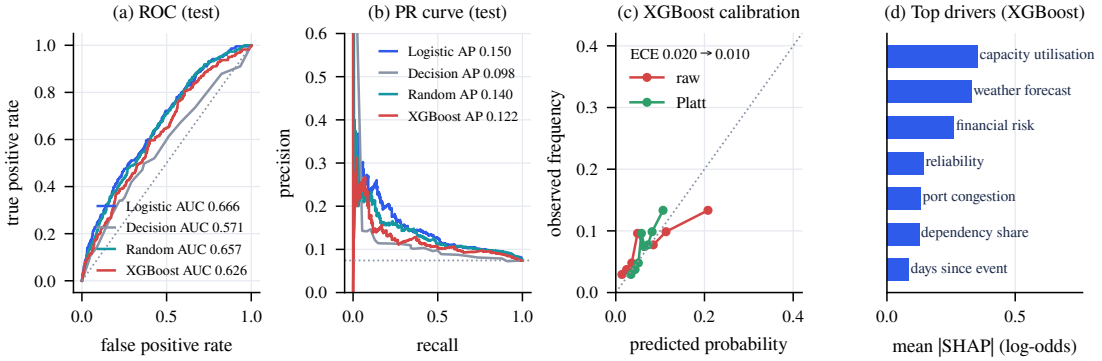


Fig. 2. E1 on the untouched test split (3,000 supplier-months, synthetic): (a) ROC and (b) precision–recall of the four calibrated models, (c) XGBoost reliability before/after Platt scaling, (d) mean absolute SHAP (SHapley Additive exPlanations) value of the top XGBoost features.

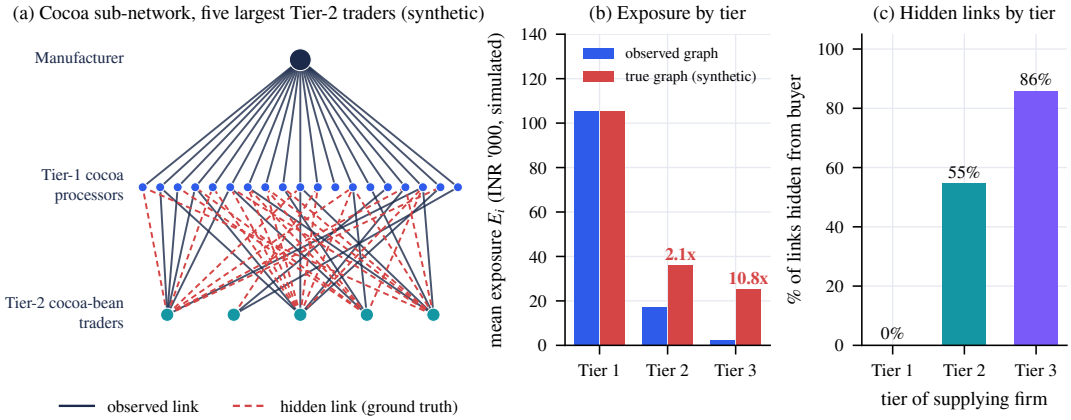


Fig. 3. The black-box gap in the synthetic network: (a) cocoa sub-network—dashed links exist in the synthetic ground truth but are hidden from the buyer; (b) mean exposure E_i (expected margin loss, Eq. (2)) per supplier on the observed vs. true graph; (c) share of hidden links by tier.

TABLE V
E2A LINK PREDICTION (SYNTHETIC, 5 SEEDS FOR LEARNED MODELS). A1: HELD-OUT OBSERVED EDGES VS. MATERIAL-COMPATIBLE NEGATIVES. A2: RANKING ALL 8,863 COMPATIBLE UNOBSERVED PAIRS AGAINST 1,039 GROUND-TRUTH HIDDEN EDGES (BASE RATE 0.117). LEARNED MODELS: MEAN±SD OVER 5 SEEDS; THE SEED-0 MLP LIST USED DOWNSTREAM HAS AP 0.220 AND P@100 0.29. ADAMIC–ADAR EQUALS COMMON NEIGHBOURS. P@100/R@500 = PRECISION OF TOP 100/RECALL OF TOP 500.

	A1 (held-out)		A2 (hidden-edge discovery)		
Method	ROC-AUC	AP	AP	P@100	R@500
Common neigh.	0.500	0.500	0.117	0.06	0.042
Pref. attach.	0.733	0.763	0.166	0.34	0.120
Node2Vec	0.638	0.609	0.146	0.24	0.084
MLP	0.826±0.017	0.849±0.012	0.227±0.010	0.32	0.141
GCN	0.807±0.020	0.837±0.010	0.176±0.007	0.25	0.109
GraphSAGE	0.737±0.009	0.784±0.012	0.168±0.005	0.29	0.102
GAT	0.796±0.005	0.811±0.018	0.177±0.008	0.26	0.103

(+0.035, p <0.001); delay-days regression adds nothing (R^2 0.27 vs. 0.28). This is a separate lateness task; it does not validate the network-level results.

D. Executable Decision Example (D1, D2)

For supplier S0065 (Tier-1) at t=2025-12-01, XGBoost gives 40.8 % raw and 15.2 % calibrated risk and E_i = INR 4.41 L; with break-even thresholds of 7.6/11.3/36.3 %, expected costs are 66.9/61.9/56.8/80.1 thousand INR, so a contingency review is chosen, and the MILP returns an optimal standby plan (INR 15.1k). This shows traceability, not that the rule improves outcomes.

E. Nine-Scenario Evaluation (RQ4a: not supported overall)

Table VI and Fig. 4 summarise the nine scenarios; Table IV gives the tests. *Severe*: A leaves 734k units unmet (4.72 Cr), B 605k (3.98 Cr) and C 39k (1.16 Cr). C saves 3.56 Cr over A (95 % CI 3.25–3.87; median 2.83 Cr) and is cheaper in 80 % of replications. *Moderate*: C saves 16.7 L over A on the mean, yet in the median replication C costs 11.8 L *more* and is cheaper in only 17 %; the significant Wilcoxon test reflects that typical shift against C, while the mean is driven by rare long outages. *Mild*: impact is mostly assumed management overhead; C costs 0.82 L more than A in every replication (small spread: only its pre-booked orders differ). H4a is therefore not supported; C is preferable only for severe shocks.

F. Value of Predicted Links (RQ4b: severe shocks only)

With the MILP fixed, predicted links cut severe impact from 3.64 to 1.16 Cr (saving 2.48 Cr, CI 2.21–2.78; Table VII),

TABLE VI

NINE-SCENARIO EVALUATION (MEANS OF 500 COMMON-RANDOM-NUMBER REPLICATIONS; SIMULATED INR ON SYNTHETIC DATA). SEVERITY = CAPACITY LOSS OF THE FIVE LARGEST TIER-2 COCOA-BEAN TRADERS FOR A NOMINAL 60 DAYS (LOG-NORMAL). POLICIES: A NONE, B RULE-BASED, C OPTIMISATION. L = LAKH, CR = CRORE; PI = 2.5–97.5TH PERCENTILE OVER REPLICATIONS; FULL REC. = REPLICATIONS WITH ZERO UNMET DEMAND; ALT. UTIL. = SHARE OF ALTERNATES' SPARE CAPACITY USED.

Scen.	Sev.	Pol.	Aff. sup.	Aff. mat.	Unmet (k units)	Inv. short. (L)	Procure. (L)	Transport (L)	Impact, Cr [95% PI]	Recov. d	Alt. util. %	Full rec. %
S1	Mil.	A	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	0	100
S2	Mil.	B	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	1	100
S3	Mil.	C	10.4	2.4	0.0	0.0	4.8	0.12	0.26 [0.01, 0.46]	101	5	100
S4	Mod.	A	17.1	3.9	47.6	2.8	0.0	0.00	0.61 [0.23, 4.54]	123	0	82
S5	Mod.	B	17.1	3.9	29.8	1.9	3.0	0.12	0.51 [0.23, 3.70]	123	16	88
S6	Mod.	C	17.1	3.7	2.6	0.2	86.2	2.95	0.45 [0.38, 0.63]	123	16	97
S7	Sev.	A	19.8	4.0	734.1	46.3	0.0	0.00	4.72 [0.23, 16.10]	127	0	14
S8	Sev.	B	19.8	4.0	605.3	38.9	18.3	0.73	3.98 [0.24, 15.19]	126	42	15
S9	Sev.	C	19.8	3.7	39.2	5.3	461.9	24.20	1.16 [0.93, 3.84]	125	23	88

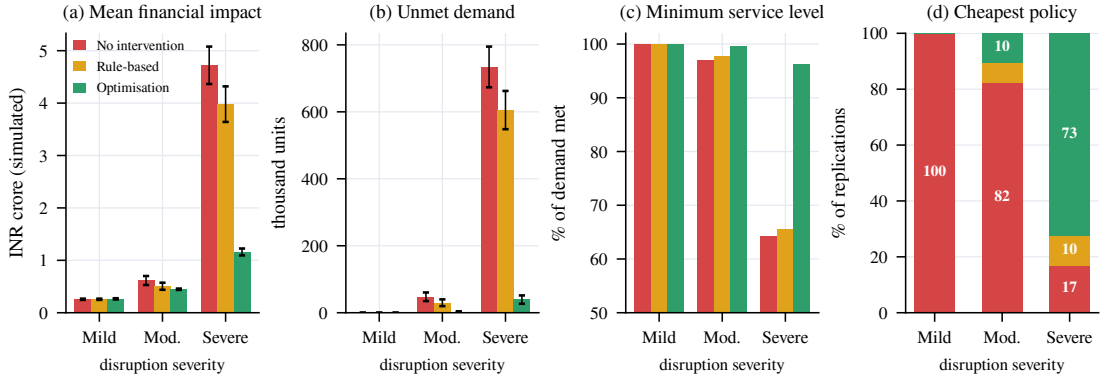


Fig. 4. Simulated outcomes on the synthetic supply network across mild, moderate and severe disruption scenarios (500 common-random-number replications): (a) mean financial impact in INR crore; (b) unmet demand in thousand units; (c) minimum service level achieved; and (d) percentage of replications in which each policy is the cheapest. Error bars in (a) and (b): 95 % CI of the mean.

TABLE VII

LINK-INFORMATION ABLATION (RQ4B). MEAN SIMULATED IMPACT, INR LAKH ± 95% CI (500 CRN REPLICATIONS); ONLY LOGIC OR KNOWN GRAPH CHANGES. LOWER BLOCK: PAIRED SAVING, MEAN/MEDIAN, HOLM-ADJUSTED WILCOXON p OVER 12 COMPARISONS.

Arm	Logic	Graph known	Mild	Moderate	Severe
A	–	–	25.7±1.2	61.5±8.6	472.1±35.7
B-obs	rule	observed	25.7±1.2	50.5±6.8	398.0±33.9
B-pred	rule	obs.+pred.	25.7±1.2	62.0±8.6	472.6±35.7
C-obs	MILP	observed	25.7±1.2	48.6±6.4	363.6±32.1
C-pred = C	MILP	obs.+pred.	26.5±1.2	44.8±1.3	115.9±6.6
C-oracle	MILP	true	25.7±1.2	38.3±2.1	120.7±8.1
Links, MILP: C-obs−C			-0.8/-0.8	+3.9/-11.7	+247.7/+148.5
		p_{Holm}	<.001	<.001	<.001
Links, rule: B-obs−B-pred			+0.0/+0.0	-11.5/-0.5	-74.6/-93.8
		p_{Holm}	.109	<.001	<.001
MILP, observed: B-obs−C-obs			-0.0/+0.0	+1.9/+0.0	+34.3/+36.4
		p_{Holm}	.004	.005	<.001

TABLE VIII

ONE-AT-A-TIME SENSITIVITY (200 CRN REPLICATIONS PER CELL). SAVING = IMPACT(A)−IMPACT(C); TIES GO TO A. KNOWN-LINK ROWS CHANGE WHAT B AND C KNOW, NOT THE TRUE NETWORK. †CHANGES THE RANDOM STREAM; COMPARE WITHIN ROW.

	Severe		Mild	
Setting	C−A saving, Cr [95% CI]	Best	Saving, L	Best
Base (200 reps)	3.28 [2.77, 3.79]	C	-0.82	A
Mgmt. overhead 0/day	3.27 [2.76, 3.78]	C	-0.82	A
Mgmt. overhead 50k/day	3.28 [2.78, 3.79]	C	-0.82	A
Shortage penalty 5%	2.58 [2.17, 3.00]	C	-0.82	A
Shortage penalty 30%	4.35 [3.66, 5.05]	C	-0.82	A
Interruption 25k/plant-day	2.94 [2.46, 3.42]	C	-0.82	A
Interruption 150k/plant-day	3.79 [3.23, 4.34]	C	-0.82	A
Nominal duration 30 d	0.21 [0.07, 0.35]	C	+0.00	A
Nominal duration 90 d	7.74 [7.08, 8.39]	C	-5.12	A
Alt. spare capacity ×0.5	2.96 [2.48, 3.43]	C	-0.81	A
Alt. spare capacity ×1.5	3.33 [2.80, 3.86]	C	-0.83	A
Alt. failure risk ×0.5†	4.20 [3.71, 4.68]	C	-0.82	A
Alt. failure risk ×2†	3.47 [2.95, 3.99]	C	-0.82	A
Known deep links −50%	3.25 [2.77, 3.74]	C	-0.82	A
Known hidden links +50%	3.23 [2.71, 3.75]	C	-0.83	A

about 88 % of C's gain over B; the MILP alone on the observed graph saves only 0.34 Cr. Predicted links match the oracle graph here (1.21 Cr). They are not universally helpful: under the moderate shock the MILP gains on the mean but loses in the median replication (11.7 L), and under the mild shock the predicted links cause the over-hedging. With the rule-based policy predicted links make outcomes worse (4.73 vs. 3.98 Cr): hard exclusion of every linked alternate leaves no usable supplier, so predicted links pay only with quantitative de-rating.

G. Assumption Sensitivity

C has the lowest severe impact in all 15 settings and A the lowest mild impact in all 15 (Table VIII). The severe saving

ranges from 2.58 to 4.35 Cr across cost, capacity, risk and visibility settings and from 0.21 to 7.74 Cr for 30- or 90-day outages: the ranking is robust, the amount is not.

Key findings (synthetic network; simulated INR)

1 Observed links understate Tier-2/3 exposure 2.1×/10.8×. **2** LR > XGBoost ≥ GNNs; calibration gains are not significant. **3** Optimisation pays only for severe shocks ⇒ a severity-contingent policy. **4** About 88 % of the severe-shock gain over the rule-based policy comes from predicted links, used with de-rating rather than exclusion.

IV. Discussion

Established (synthetic network, assumed costs): predicted links, used by a MILP, lower simulated severe-shock impact. **Not established:** real-world savings, real-network link quality and decision benefit of the D1 rule. **Threats to validity.** *Internal:* the authors wrote the generator, so methods matched to it may look better than on real data. *Construct:* impact uses illustrative unit costs. *External:* the network-level results are synthetic; DataCo and SupplyGraph test components only. *Statistical:* E1 has 223 test positives; scenario CIs ignore parameter uncertainty, which Table VIII explores one factor at a time only.

V. Research Integrity, AI Use and Reproducibility

Integrity. No choice was tuned on test data. Scripts rebuild every table, figure and quoted number from saved results; 73 logged checks recompute splits, metrics, CRN integrity and impact components, and an independent audit recomputes 87 quoted numbers from raw data, saved models and per-replication results (`experiments/`). Disclosed deviations: (i) three simulator corrections after the first scenario runs (a storage cap that discarded deliveries, C excluding rather than de-rating alternates, 180-day horizon); (ii) tests, ablation and sensitivity analysis were added after reviews of the first draft. **AI assistance disclosure.** Generative AI (Claude Code, Anthropic; 12–13 Nov 2026) was used extensively as a development and writing assistant: under the authors' direction it generated most of the code, executed the experiments and drafted the text, and all repository commits were produced in AI-assisted sessions. The authors defined the problem, scope, datasets and evaluation requirements, reviewed the report and directed its revisions after faculty review. All authors take responsibility for the submitted work. The Gemini API is separately a functional component of the deployed assistant and was not used to write this report. **Reproducibility.** Code, data and logs: https://github.com/SamarthKamat/AIML-PROJECT (public). Application (React, Flask, 31 API routes, SQLite in deployment, MySQL supported, 35 tests passing): https://supply-chain-black-box.onrender.com. `run_all.sh` downloads both real datasets at pinned commits (DataCo SHA-256 checked) and runs every stage.

VI. Conclusion

RQ1 partly supported, RQ2 supported, RQ3 not supported, RQ4a not supported overall and RQ4b supported for severe shocks only (Table II). Next: validate on real supplier maps (bills of lading, UN Comtrade cocoa flows) and learn severity-contingent, rolling-horizon policies.

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